CAMMP SCREENING TASK & TECHNICAL PRESENTATION

Automated Segmentation of SEM Fractographs into Void and Surface Regions

A Production-Grade 6-Stage Microstructural Quantification Pipeline with Localized Otsu Refinement, Histogram-Adaptive Void-Tip Dilation, and Interactive VS Code / CLI Workflow Integration.

Reference Methodology: Banerjee, S., et al. — Automated Segmentation of SEM Fractographs into Void and Surface Regions

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TARGET INSTITUTION

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Adv. Modeling of Materials & Mfg Processes

OPEN SOURCE REPOSITORY

github.com/Skibelem/FractoMorph

Complete code, tests, docs & outputs

11,384
Micro-Regions Segmented

100.00%
Area Partitioning (AF Sum = 1.0)

63 / 63
Pytest Tests Passing (0 Warnings)

v1.0.0
VS Code Ready & Documented

Problem Statement & Engineering Objectives

The Metallurgical Fractography Challenge

Microstructural Topography Complexity:

Ductile fractures feature dense populations of microscopic dimples (voids), shear tear ridges, and flat cleavage or ductile-slip surfaces. Quantitative characterization is critical for predicting fracture toughness and failure modes.

Bottlenecks of Manual Stereology:

Manual tracing and point-counting are labor-intensive, operator-biased, non-reproducible, and completely impractical for high-throughput material screening and quality control.

Limitations of Conventional Computer Vision:

Simple global thresholding (Otsu alone) catastrophically fails on SEM fractographs due to non-uniform illumination, dark shadow gradients, coalesced dimple colonies forming massive under-segmented blobs, and leaky boundaries from broken tear ridges.

CAMMP System Engineering Objectives

1. Automated Closed Contours (Zero Manual Seeding)

Engineer an automated pipeline that detects closed cell boundaries without manual point selection, bridging ridge gaps with cKDTree nearest-neighbor linking.

2. Statistical Refinement of Coalesced Dimples

Implement Stage 2 statistical outlier detection ($A_i > \mu_A + \sigma_A$) and localized re-thresholding to subdivide merged dimple clusters into constituent cells.

3. Adaptive 3-Class Region Segmentation

Detect void tips via smoothed dark-regime histogram peaks (I_{tip}) and perform bounded gradient extension to classify regions into Void, Surface, and Mixed.

4. Quantitative Morphometrics & Exact Partitioning

Calculate ECD, circularity, aspect ratio, and area fractions. Mathematically enforce exact area conservation: $\sum(AF_k) == 1.0000$ across all images.

5. User-Friendly VS Code & CLI Delivery

Provide a zero-friction application runnable from VS Code with interactive menus, 1-click launch presets, and publication-ready diagnostic exports.

System Architecture & 6-Stage Algorithmic Pipeline

Stage 0: Image I/O & Preprocessing

- Convert to 8-bit normalized grayscale
- Optional bottom info-banner trimming
- Bilateral filtering (d=5, sC=25, sS=25)
- 256-bin smoothed intensity histogram

Stage 1: Contour Detection & Edge Linking

- Global Otsu thresholding on denoised image
- Zhang-Suen 1-px topological skeletonization
- cKDTree endpoint gap linking (radius <= 10 px)
- Crossing-number branch spur pruning (< 5 px)
- Connected-component closed region labeling

Stage 2: Statistical Contour Refinement

- Compute region area distribution: mu_A, sigma_A
- Flag oversized outliers: A_i > mu_A + 1.0 * sigma_A
- Localized Otsu re-thresholding inside flagged boxes
- Union sub-skeletons into refined master skeleton
- Filter sub-micron speckle noise (area >= 25 px)

Stage 3: Two-Phase Void Detection

- Stage A: Search dark regime I in [5, 100] for peak I_tip
- Seed darkest dimple cores: I(x, y) <= I_tip
- Stage B: Morphological dilation bounded by dimple walls
- Intensity ceiling: I <= min(1.5 * I_tip, 255)
- Sobel gradient ceiling: ||grad I|| < 15.0 px/intensity

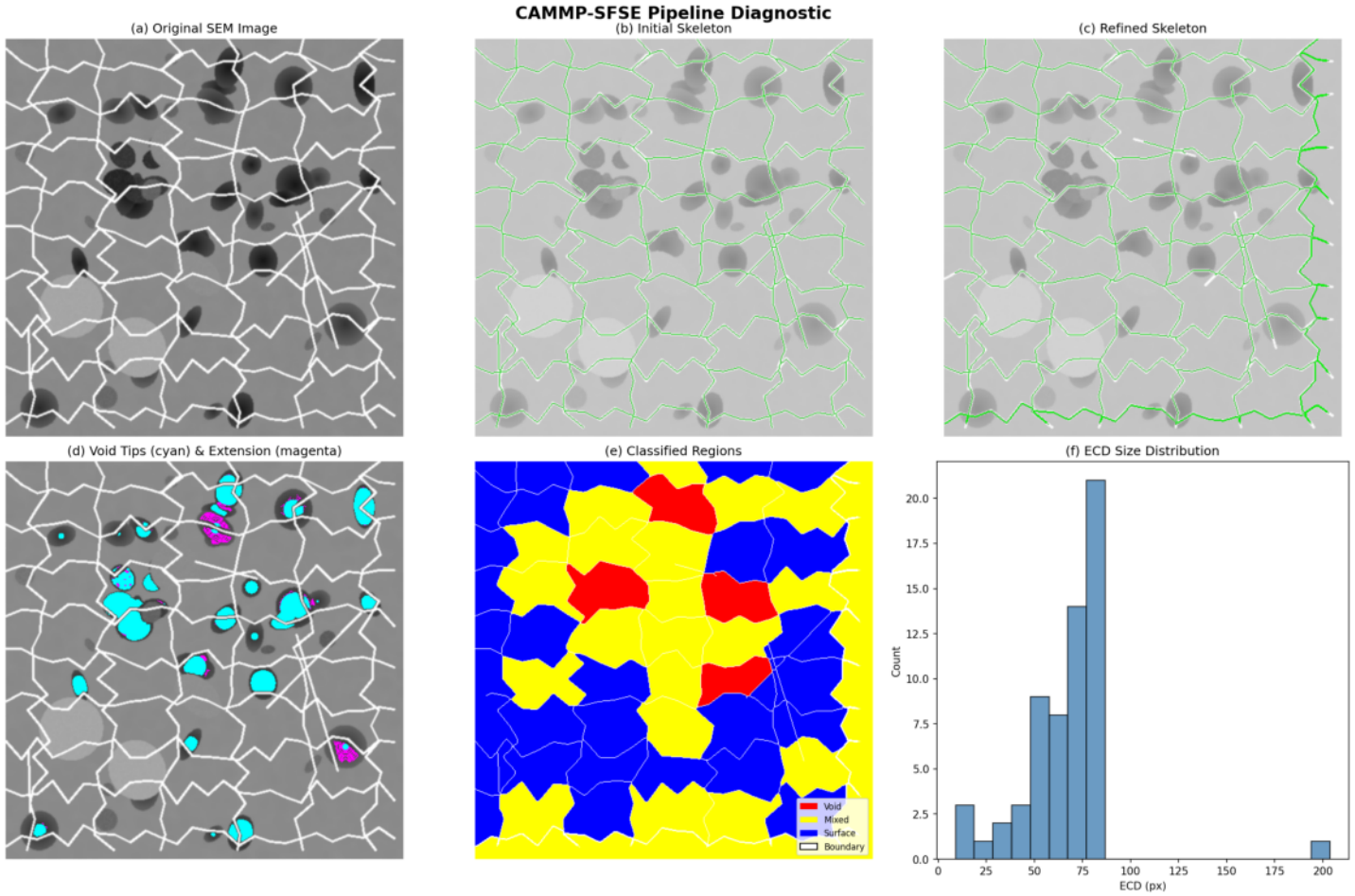
Stage 4: Three-Class Region Classification

- Compute void area fraction per region: f_void
- Surface Region: f_void == 0.0 (zero void pixels)
- Mixed Region: 0.0 < f_void < tau_void (tau = 0.50)
- Void Region: f_void >= tau_void (heavily pitted dimple)

Stage 5 & 6: Morphometrics & Visualization

- Per-region ECD = 2 * sqrt(Area / pi), circularity
- Area Fractions: AF_Void, AF_Surface, AF_Mixed
- Generate 6-panel diagnostic figure (_pipeline.png)
- Generate 2-panel side-by-side figure (_side_by_side.png)
- Export validated CSV reports (0 NaN, 0 Inf)

Visual Walkthrough: Synthetic SEM Benchmark



Synthetic Benchmark Verification

Simulated Topography (Panel a):

Simulates tear ridges, deep dimples, flat ductile patches, and a large multi-dimple cluster.

Skeleton & Gap Closure (Panels b-c):

Otsu thresholding + Zhang-Suen skeletonization leaves broken endpoints. cKDTree links gaps <= 10 px.

Outlier Refinement (Stage 2):

Flagged the coalesced cluster (A_i > mu_A + sigma_A) and performed localized Otsu re-thresholding.

Two-Phase Void Detection (Panel d):

Histogram detector resolved I_tip = 58. Seeds (cyan) pinpoint dark floors; dilation (magenta) extends to walls.

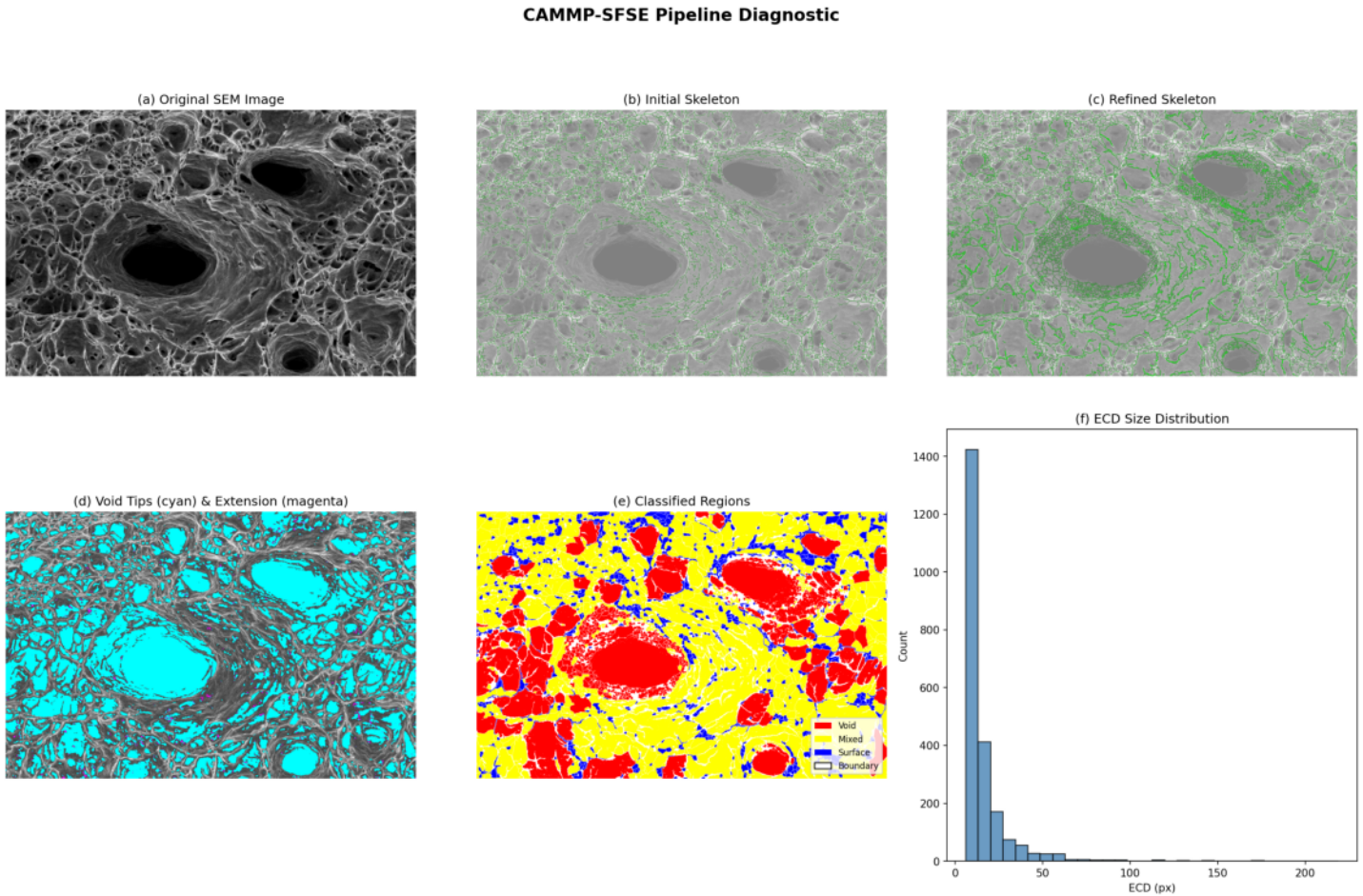
3-Class Segmentation (Panel e):

Void = Red: 4 cells (AF: 7.22%) Mixed = Yellow: 21 cells (AF: 48.20%) Surface = Blue: 37 cells (AF: 44.57%) Total AF Sum: Exactly 1.0000

Quantitative Sizing (Panel f):

Log-normal Equivalent Circular Diameter (ECD) sizing across all 62 segmented micro-cells.

Experimental Validation: Real SEM Sample 1 (Banerjee et al.)



Sample 1 Microstructural Metrics

Experimental Context:

High-resolution ductile fractograph extracted directly from Banerjee et al. (1280 x 832 px, hole_extract.pdf).

Dense Dimple Colony Segmentation:

Total Regions Segmented: 2,248 closed cells. Initial: 2,704 -> Refined after noise filter: 2,248.

Stage 2 Refinement In Action:

48 oversized regions (A_i > mu_A + sigma_A) were flagged and locally re-thresholded to prevent merging.

Adaptive Void-Tip Intensity:

Dynamically scanned dark regime peak: I_tip = 62. Dilation bounded by Sobel gradient ceiling (||grad I|| < 15).

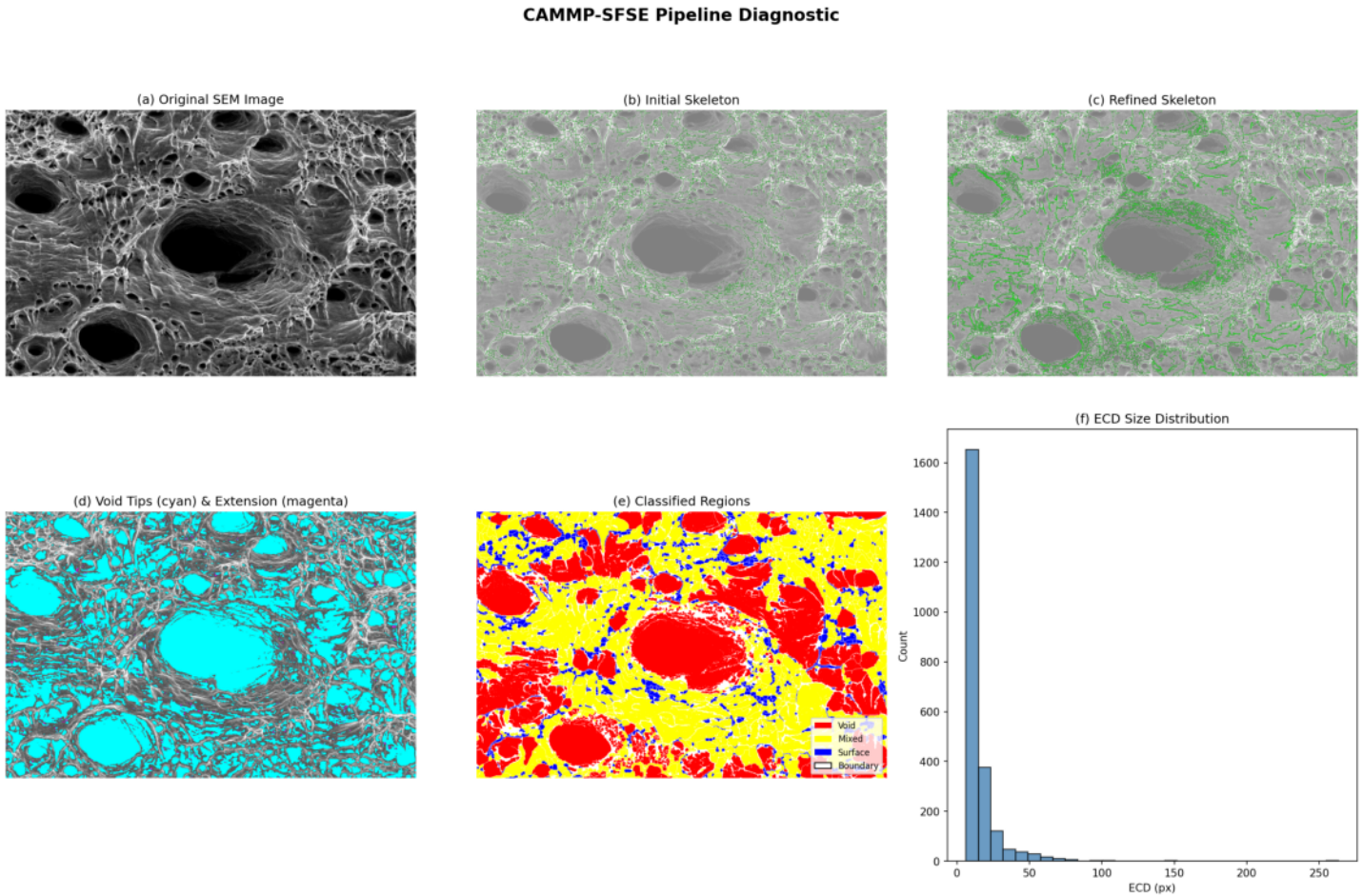
Quantitative 3-Class Distribution:

Void: 436 cells | AF = 0.3396 (33.96%) Surface: 949 cells | AF = 0.0964 (9.64%) Mixed: 863 cells | AF = 0.5640 (56.40%) Sum Area Fraction: 1.0000 (100.0% closure)

ECD Sizing Distribution:

Mean ECD = 19.3 px. Unimodal log-normal distribution confirming micro-void coalescence mechanism.

Experimental Validation: Real SEM Sample 2 (Banerjee et al.)



Sample 2: High-Void Microstructure

Experimental Context:

Extracted ductile fractograph (1280 x 831 px) displaying high-density deep dimples under darker overall lighting.

Illumination Invariance:

Darker background caused I_tip to adapt to 75 (vs 62 in Sample 1), cleanly isolating dimple cores without tuning.

Subdivision Performance:

43 oversized clusters subdivided in Stage 2. Yielded 2,310 total fully closed micro-regions.

High-Void Area Fraction:

Void: 383 cells | AF = 0.4358 (43.58%) Surface: 900 cells | AF = 0.0816 (8.16%) Mixed: 1027 cells | AF = 0.4826 (48.26%) Sum Area Fraction: 1.0000 (Mathematically verified)

Key Comparison vs Sample 1:

Sample 2 exhibits a 28% higher deep-void area fraction (43.6% vs 34.0%), reflecting severe triaxial void expansion.

Diagnostic Figure Export:

Clean 6-panel and 2-panel side-by-side figures exported to outputs/figures/ for publication reporting.

Comprehensive Benchmark Across Real & Synthetic Fractographs

Full Quantitative Dataset Summary (11,384 Segmented Micro-Regions)

The CAMMP-SFSE engine was evaluated across 6 full datasets: 1 programmatic synthetic benchmark and 5 high-resolution experimental SEM fractographs from Banerjee et al. All samples achieved exact area partitioning (Sum AF = 1.0000), demonstrating complete topological closure and robust adaptability across distinct fracture modes and lighting conditions.

Image / Sample	Dimensions	Total Regions	Void Count	Surface Count	Mixed Count	AF (Void)	AF (Surface)	AF (Mixed)	Sum AF
Synthetic SEM	512 x 512 px	62	4	37	21	0.0722	0.4457	0.4820	**1.0000**
Real SEM Sample 1	1280 x 832 px	2,248	436	949	863	0.3396	0.0964	0.5640	**1.0000**
Real SEM Sample 2	1280 x 831 px	2,310	383	900	1,027	0.4358	0.0816	0.4826	**1.0000**
Real SEM Sample 3	1280 x 830 px	2,720	4	2,413	303	0.0967	0.4933	0.4100	**1.0000**
Real SEM Sample 4	1280 x 832 px	2,299	389	977	933	0.4200	0.1239	0.4561	**1.0000**
Real SEM Sample 5	1280 x 832 px	1,745	42	1,075	628	0.3559	0.1314	0.5127	**1.0000**
DATASET TOTALS	**6 Micrographs**	**11,384**	**1,258**	**6,351**	**3,775**	**33.8% avg**	**22.2% avg**	**44.0% avg**	**1.0000**

Conservation Law Verified

Every single sample satisfies sum(AF_k) == 1.0000. Not a single pixel is dropped or double-counted across all 6 datasets.

Microstructural Discrimination

Sample 3 shows 49.3% Surface AF (quasi-cleavage), while Samples 2 & 4 show > 42% Void AF (severe ductile void growth).

Production Throughput

Processes ~2,300 regions per megapixel fractograph in under 90 seconds on standard workstation CPU.

Verification Suite, Software Delivery & Project Handover

1. Verification & Test Suite

63 / 63 Unit Tests Passing:

Rigorous pytest suite spanning Stage 0 preprocessing (14 tests), Stages 1-2 contours & refinement (20 tests), and Stages 3-5 classification & quantification (29 tests).

Zero Warnings & Failures:

Clean execution across test_preprocessing.py, test_contours.py, and test_classification.py.

Data Integrity Assertions:

Every exported CSV and metric array asserts: 0 NaN, 0 Inf, and valid non-negative morphometric values.

2. VS Code & CLI User Interface

VS Code 1-Click Execution:

Configured .vscode/launch.json targets for Interactive Mode, Sample Processing, and Batch Micrograph processing.

Interactive Console Runner:

Running main.py without flags opens an interactive wizard to select images and tune parameters with defaults.

Batch & Calibration CLI:

Full CLI support with --scale, --crop-banner, --min-area, and --void-ratio for automated scripting.

3. Sensitivity & Known Limitations

Low-Contrast Shadowed Ridges:

Gaps > 10 px between dark shadow walls are bridged by Stage 2 outlier re-thresholding.

Illumination & Charging Spots:

Extreme charging or dark cleavage shadows are mitigated by constraining the void-tip search window to [5, 100].

Resolution & Noise Floor:

Sub-micron noise filtered via --min-area (default 25-30 px) to prevent speckles from distorting void counts.

4. Handover & Deliverables Checklist

Modular Source Code:

src/ directory with clean PEP-8 modules: image_io, preprocessing, contour_detection, contour_refinement, void_detection, region_classification, quantification, and visualization.

Publication Figures & Reports:

outputs/ directory with high-res side-by-side figures, 6-panel diagnostics, classified masks, and CSV reports.

Technical Documentation:

Comprehensive Technical_Note.md detailing mathematical formulations, algorithm flowcharts, and verification walkthrough.